Which hand?

Detecting left- vs right-hand nose touches from smartphone motion sensors (brief version)

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Abstract

The task is to predict from a smartphone's motion sensors whether the left or the right hand brought it from arm's length to the nose, as a calibrated probability $P(\text{right})$, for persons the model has never seen. The data are 598 instructed trials of 20 persons, and every evaluation leaves one person out. As the mirror symmetry of the body predicts, the phone's early rotation about gravity changes sign with the hand. Among 17 candidates, a pre-specified rule chose a one-dimensional convolutional network; the nested estimate of this selection, with the candidates fixed, is an accuracy of 88.5 % [84.1, 92.3]. With its in-fold Platt calibrator its probabilities are consistent with calibration, and three attempts raise its accuracy to 96.4 % (exploratory). Without a start cue, where the final model has not yet run, a simpler reference classifier shows a hand for 76.4 % of the recordings and is right on 79.9 % of them (forced choice 74.6 %; exploratory).

1 The question

The task, set by the principal investigator (PI), is to predict which hand brings a smartphone to the nose from the motion sensors that the browser delivers to a web page at about 60 Hz [1].

The answer is a percentage, $P(\text{right})$: the probability that the right hand moved the phone. It must be calibrated (of the touches given 80 %, about 80 % should be right-hand touches) and hold for persons the model has never seen.

The two hands make mirror-image movements, so simple descriptors such as speed and duration are much the same for both; persons, phones and grips add their own variation. In production there is also no start cue: the page must find the approach in a continuous stream and decide when it has seen enough.

Figure 1 shows the pipeline. Every evaluation leaves one person out (LOSO), because splits by trial overstate the accuracy for new persons [2].

This version is preliminary: the analysis ran in lanes, one per model family or analysis step, and the final-model lane, which selects, calibrates and exports the model, is a first pass, not yet independently audited. The full report gives every decision with its evidence (Appendices A to I).

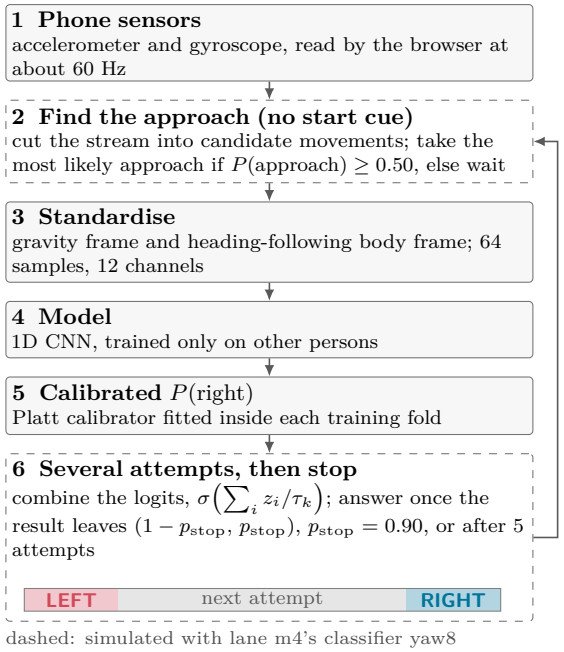


Figure 1: From the phone's sensors to the answer shown. The dashed steps, which production needs without a start cue, were simulated in lane m4 with a simpler reference classifier, yaw8 (Section 3), not yet with the final model (Section 4).

2 Why the hand is detectable

The data come from 20 students (2 left-handed; 12 on Android, 8 on iOS), who followed the prompts of the NoseTouch web app for the hand, the grip (upright or sideways) and the speed of each trial.[1] A pre-specified exclusion cascade keeps 598 instructed trials and 20 free-choice ones of the 878 recorded attempts.

[1] A video of a data-collection session: https://drive.google.com/file/d/1V9PF623HkJ8MjR6loJhUHnDkCx60W-Jl/view?usp=sharing.

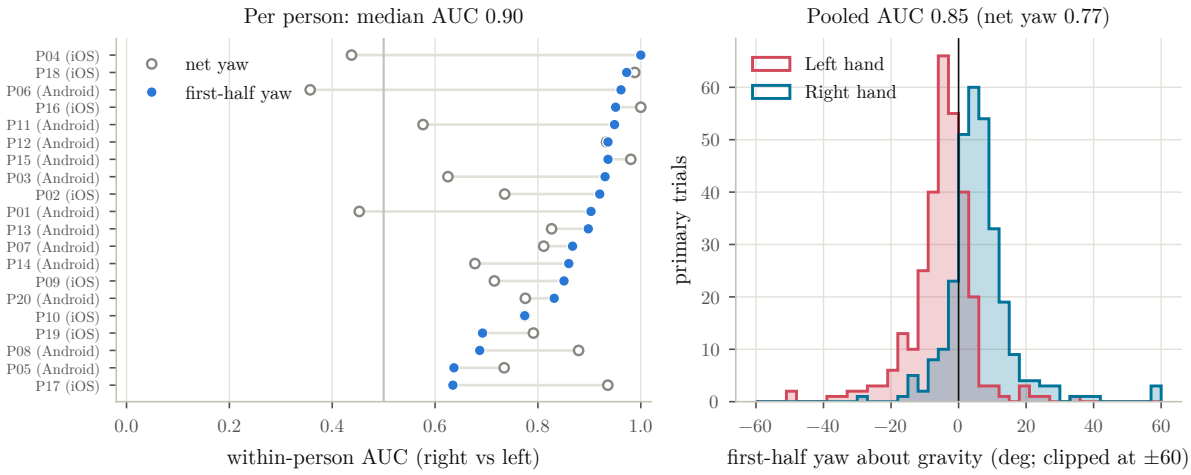


Figure 2: The yaw about gravity accumulated over the first half of the approach separates the hands (left: AUC within each person; right: the pooled distributions). The net yaw over the whole approach (grey) has a lower pooled AUC, 0.77, not tested in a paired comparison.

In the free-choice trial each person chose the hand and reported it afterwards; these natural-use trials, all right-hand, are scored but never used for training (full report, Section 2).

Mirroring the body across its midline swaps the hands. It flips the sign of chiral (mirror-odd) quantities, such as the yaw rate about gravity, and leaves mirror-even ones, such as speed and duration, unchanged.

If the two hands move as mirror images, only chiral quantities can carry the hand. The yaw rate about gravity is also unchanged by any grip, because a grip only rotates the phone's axes.

Early in the approach the forearm swings toward the body's midline, so, seen from above, the phone turns counter-clockwise in the right hand and clockwise in the left.

The yaw about gravity accumulated over the first half of the approach separates the hands with a pooled area under the ROC curve (AUC) of 0.85, and within persons the AUC is above 0.5 for 20 of 20 persons (Figure 2). Its sign alone, with no training, classifies 79 % of the trials.

The models therefore get time-resolved channels in a gravity frame, and a body frame that follows the phone's heading adds channels that do not depend on an upright or sideways grip (a screen-away hold needs a separate check, Section 3).

3 Results

These results use the approach window that the app stored during recording, with all tuning nested inside the training persons of each LOSO fold. Every 95 % confidence interval (CI) is a person-cluster bootstrap interval, from 2000 resamples of persons [3], and each p-value names its family of comparisons and its adjustment.

Four model families were built: rules and interpretable models on physics features, classical machine learning on engineered features, time-frequency and deep sequence models, and a reference floor fixed before any model was built, yaw8 (a mirror-augmented logistic regression on the cumulative yaw at 8 instants).

A rule fixed before the comparison chose the final model among 17 candidates, each scored as it would ship (full report, Appendix F). It keeps the leaders that paired tests cannot separate from the lowest log loss, drops those that fail on a grip absent from training, on natural use, or with the screen facing away even after a rest-pose check, and takes the one easiest to run in a browser.

The rule chose a one-dimensional convolutional network (1D CNN) on 12 gravity-frame and body-frame channels: the final model. Rerunning the whole selection inside every outer fold, on the training persons only and with the candidates fixed, gives the headline: a nested, selection-aware accuracy of 88.5 % [84.1, 92.3] and an AUC of 0.945 [0.906, 0.975] (Table 1).

The final model's own LOSO accuracy, 89.3 % [85.0, 93.1], is exploratory: it was the best of 16 configurations compared on the outer folds, then of 17 candidates, and its optimism over the nested estimate is +0.8 [−0.2, +2.3] points. The results below describe this shipped model and share its exploratory status.

The final model is +11.7 [+8.0, +15.6] points more accurate than the reference floor yaw8, which reaches 77.9 % [73.1, 82.9] (Holm $p < 0.001$ over the 16 comparisons with the reference).

Accuracy varies by person: 3 of 20 persons score below 80 %, the lowest 61.8 %, and 24 of the 64 errors are confident (below 0.2 for the true hand), clustered in a few persons.

The body frame limits the cost of a grip absent from training: trained on upright grips only, the final model changes by −4.1 [−7.0, −1.4] points on the sideways grips (Holm $p = 0.188$ over the 17 grip tests).

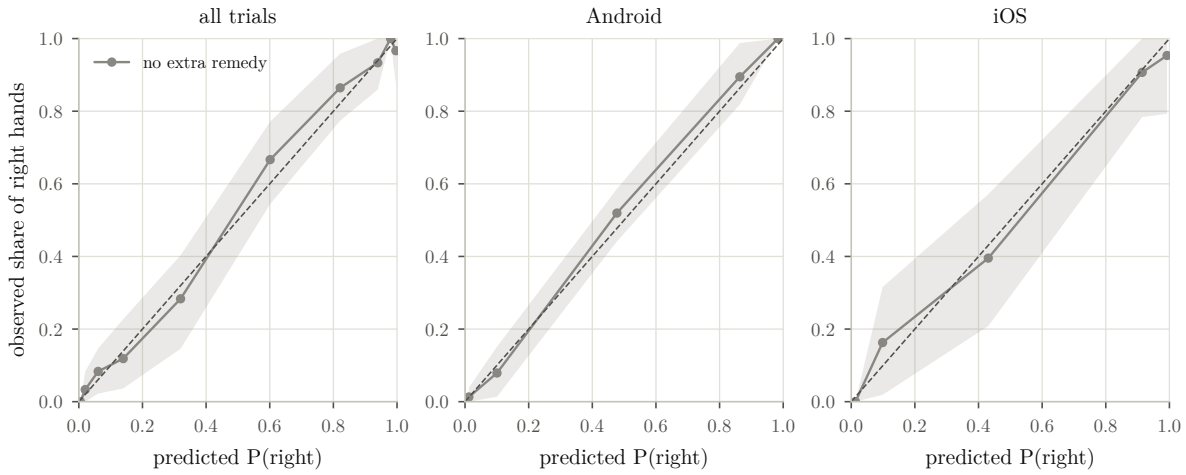


Figure 3: Reliability of the final model with its in-fold Platt calibrator, over all trials and per platform (equal-mass bins; person-bootstrap 95 % bands; dashed: perfect calibration). The percentage shown is consistent with calibration.

Table 1: Key results (95 % person-bootstrap CIs in brackets). The middle four rows describe the final model as shipped, an exploratory selection.

Question	Result
Unseen persons	nested estimate: accuracy 88.5 % [84.1, 92.3], AUC 0.945 [0.906, 0.975]
The percentage	calibration error (ECE) 0.026 (chance: up to 0.042); slope 0.97 [0.69, 1.50], intercept +0.07 [−0.25, +0.39]
Three attempts	accuracy 96.4 % [93.2, 98.9]
Unseen grip	accuracy change on sideways grips when trained on upright ones: −4.1 [−7.0, −1.4] points
Natural use	right-hand recall 85.0 %[a]
No start cue	yaw8, not the final model[b]: a hand shown for 76.4 %, right on 79.9 % of those; forced choice 74.6 %

[a] All free-choice trials are right-hand, so a constant right answer would score 100 %. [b] Lane m4's reference classifier; exploratory, the best of 22 cue-free pipelines.

The leaders that read device-frame channels change by −19.8 to −31.0 points (each Holm $p \le 0.014$ over the same 17 tests), which removed them from the selection.

A phone held with the screen facing away defeats every model that reads the screen-relative frame: on synthesised screen-away twins the final model falls to 28.6 %. An exported check of the rest pose before the movement, with an un-flip, brings it back to 86.6 %.

Deep networks are often over-confident [4]; with a Platt calibrator [5] fitted inside each training fold, the final model's probabilities are consistent with calibration (Figure 3).

Its expected calibration error (ECE), 0.026, is inside the range that a perfectly calibrated model shows by chance (up to 0.042), with slope 0.97 [0.69, 1.50] and intercept +0.07 [−0.25, +0.39] (ideal: 1 and 0).

No further remedy helps: Platt, temperature, isotonic and beta calibration, fitted inside the folds on top of it, each raise the log loss slightly, none detectably (Holm $p = 1.000$ for each, over the four remedies).

With $k = 3$ attempts of one person and hand, the accuracy of the mean $P(\text{right})$ rises to 96.4 % [93.2, 98.9] (19 persons with at least 5 trials of each hand).

The displayed aggregate, $P(\text{right})_k = \sigma\big(\sum_i z_i/\tau_k\big)$, combines the logits z_i of the attempts with a scale τ_k fitted inside each fold (σ: the logistic function). It gives nearly the same answers and is neither over- nor under-confident on average (calibration in the large only).

On the 20 free-choice trials the right-hand recall is 85.0 % (17 correct), with no detectable shortfall against the same persons' instructed trials (Holm $p = 1.000$ over the 17 candidates). All free-choice trials are right-hand so far, so a constant right answer would score 100 % and this recall says nothing about the left hand.

4 Without a start cue

In production the page receives a continuous stream: it must find the approach, reject other movements and decide when to answer. A simulation on the 598 full recordings (lane m4) ran this pipeline with its own reference classifier, yaw8, so its numbers measure what detection costs, not how the final model performs without a cue (full report, Appendix G).

The production detector cuts the stream into candidate movements and scores each with a classifier that does not use the hand. The segment it hands to yaw8 is the approach in 67 % of the recordings, and it rejects 17 % (Figure 4).

An abstention, a rejected recording or an output of exactly 0.5, is never a right-hand answer. The pipeline shows a hand for 76.4 % of the recordings and is right on 79.9 % of those (forced choice 74.6 %); it is exploratory, the best of 22 pipelines compared on the outer folds.

Against yaw8 on the approach windows that the app stored during recording, detection changes the accuracy, with half credit for abstentions, by −5.1 [−7.6,

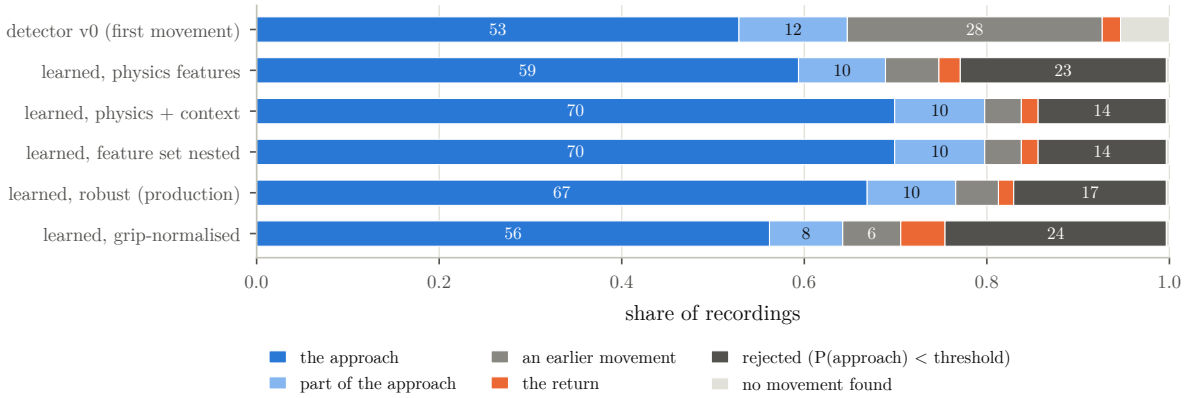


Figure 4: What each cue-free detector hands to the hand classifier on the 598 full recordings (percentages; the other learned detectors are variants compared in the full report). The production detector scores an earlier movement in 5 % of them, v0 (the first movement after a fixed stillness) in 28 %.

−2.7] points (Holm $p = 0.047$ over 26 comparisons).

In synthesised multi-attempt sessions only 77 % of the accepted attempts are approaches, and the others cancel most of the gain of combining attempts. The stopping rule of Figure 1 is post hoc and exploratory: its aggregation scale was added after the held-out sessions had shown this problem.

The detector rejects free-choice approaches more often than instructed ones (9 of 20, 45 %, against 17 %; Poisson-binomial $p = 0.0015$, unadjusted, a single pre-planned test), so it must learn them before production.

Three instructions follow: hold the phone with the screen facing you, keep still for 1 s before the movement, and make several attempts. The detector relies on that stillness: with none, the accuracy with half credit falls from 73.1 % (1 s kept) to 56.4 %.

5 Discussion

The hand that brings a phone to the nose can be read from the phone's motion sensors for persons never seen, and the signal is physical: early in the approach the phone turns about gravity in opposite senses for the two hands.

With its in-fold Platt calibrator the final model's percentage is consistent with calibration, and its body-frame input limits the change on an unseen grip to −4.1 points.

The main risk lies before the classifier: finding the approach without a cue costs accuracy, and free-choice movements are often rejected.

The data are limited: 20 students aged 19 to 29, of whom 8 used iOS and 2 are left-handed (accuracy 96.4 % and 88.0 %; no CI for two persons), and free-choice trials of the right hand only. The cue-free numbers come from yaw8, not the final model, and screen-away holds and multi-attempt sessions were synthesised, not recorded.

The nested estimate treats the 17 candidates as fixed: it does not remove the optimism of the lanes' own outer-fold choices (7 candidates; the final model the best of 16) or of the body-frame input, designed after outer-fold results had been seen.

The learning curve suggests that more persons of the same kind would add little average accuracy (89.6 % extrapolated to 40 persons, against 89.3 % with 19 training persons); new persons are needed above all for the groups the data lack.

Next steps: run the final model behind the cue-free detector, teach the detector free-choice approaches and build the inference page from the exported model (1585 kB, about 20.05 ms per prediction). Then record left-handed persons and left-hand free choice, close the open audits and let the PI test the page prospectively on new persons.

References

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